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# Dynamic relationship and pleiotropic loci of attention deficit hyperactivity disorder with sleep traits

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Attention deficit hyperactivity disorder (ADHD) is a common neurodevelopmental disorder modulated by genetic and environmental factors. Approximately 20–55% of individuals with ADHD experience concurrent sleep disturbances, yet the dynamic relationship between the two and their shared genetic mechanisms remain unclear. This study analyzed 11,288 participants (52.1% male) from the Adolescent Brain Cognitive Development Study using the Sleep Disturbance Scale for Children. We found a robust positive correlation between ADHD and sleep traits, though ADHD medication weakened this relationship. Latent growth curve modeling revealed that ADHD symptoms significantly impacted sleep traits with a lag effect. Polygenic risk score analyses showed significant but very small associations (explained variance ~0.28–0.30%). Pleiotropic analyses identified two loci associated with ADHD and excessive somnolence (implicating 30 genes), and cross-cohort comparisons with UK Biobank insomnia data uncovered 12 additional pleiotropic loci (98 genes) enriched in brain tissue. Functional enrichment analysis highlighted excitatory and inhibitory neuronal subtypes in visual and frontal cortices, suggesting roles in synaptic transmission and neural circuit development. Protein-protein interaction network analysis identified 10 hub genes involved in cell growth regulation and neurodevelopmental processes. This study demonstrates robust ADHD-sleep associations, with medication moderating this link, and provides preliminary insights into their shared genetic architecture.

*Translational Psychiatry* (2026) 16:423 ; https://doi.org/10.1038/s41398-026-04166-4

## INTRODUCTION

Attention deficit hyperactivity disorder (ADHD) is a common neurodevelopmental disorder in children [1], characterized by symptoms such as inattention, hyperactivity, and impulsivity. It typically emerges in childhood and can continue into adolescence and adulthood [2]. Globally, epidemiological data indicate that ADHD affects approximately 5% of children worldwide [1] and 6.5% of children in China [3]. The disorder is associated with significant challenges in daily functioning, disciplinary behavior, substance use, mental disorders, and sleep disturbances [4]. Meta-analyses [5–7] indicate that children with ADHD report significantly more subjective sleep problems than non-ADHD controls and also exhibit some objective sleep alterations, as measured by actigraphy or polysomnography. Furthermore, a meta-analysis by Cortese et al. demonstrated that these alterations are not attributable to comorbid psychiatric disorders or ADHD medication [5]. Specifically, children with ADHD often struggle with insomnia, excessive daytime sleepiness, and delayed sleep phase syndrome [8]. They also tend to exhibit higher rates of snoring, nighttime awakenings, and restlessness compared to their peers without ADHD [9]. On the other hand, sleep problems can impair cognitive functions such as attention, memory, language skills, emotional regulation, and behavioral control, potentially worsening ADHD symptoms [10]. Interestingly, research has shown that improving sleep quality can lead to a reduction in symptoms of inattention and hyperactivity symptoms [11]. Therefore, understanding the dynamic relationship between ADHD symptoms and sleep patterns is essential, as it has important implications for both clinical practice and public health.

In recent years, numerous observational studies have explored the relationship between sleep traits and the clinical diagnosis or symptoms of ADHD at the behavioral level. Cross-sectional studies have shown that the severity of ADHD symptom is associated with sleep disturbances throughout the lifespan [12–15]. For example, one study investigating the link between early-life sleep and ADHD in adolescents found that sleep disturbances may be a stronger predictor of ADHD than sleep duration or sleep patterns over time [16]. Longitudinally, Shen et al. found that ADHD symptoms were associated with sleep disturbances across one or more follow-up assessments [17]. In another longitudinal study, sleep disturbances at an average age of 13 were found to predict increased ADHD symptoms 5 years later [18]. However, much of these research relied on cross-sectional or case-control designs [19], and while some longitudinal studies exist, they often suffer from limitations such as limited follow-up time points [17] or small sample sizes [18]. Building on previous findings, we recognize that understanding the relationship between ADHD and sleep must extend beyond the group level to incorporate long-term

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Received: 29 December 2025 Revised: 8 May 2026 Accepted: 3 June 2026
Published online: 15 June 2026

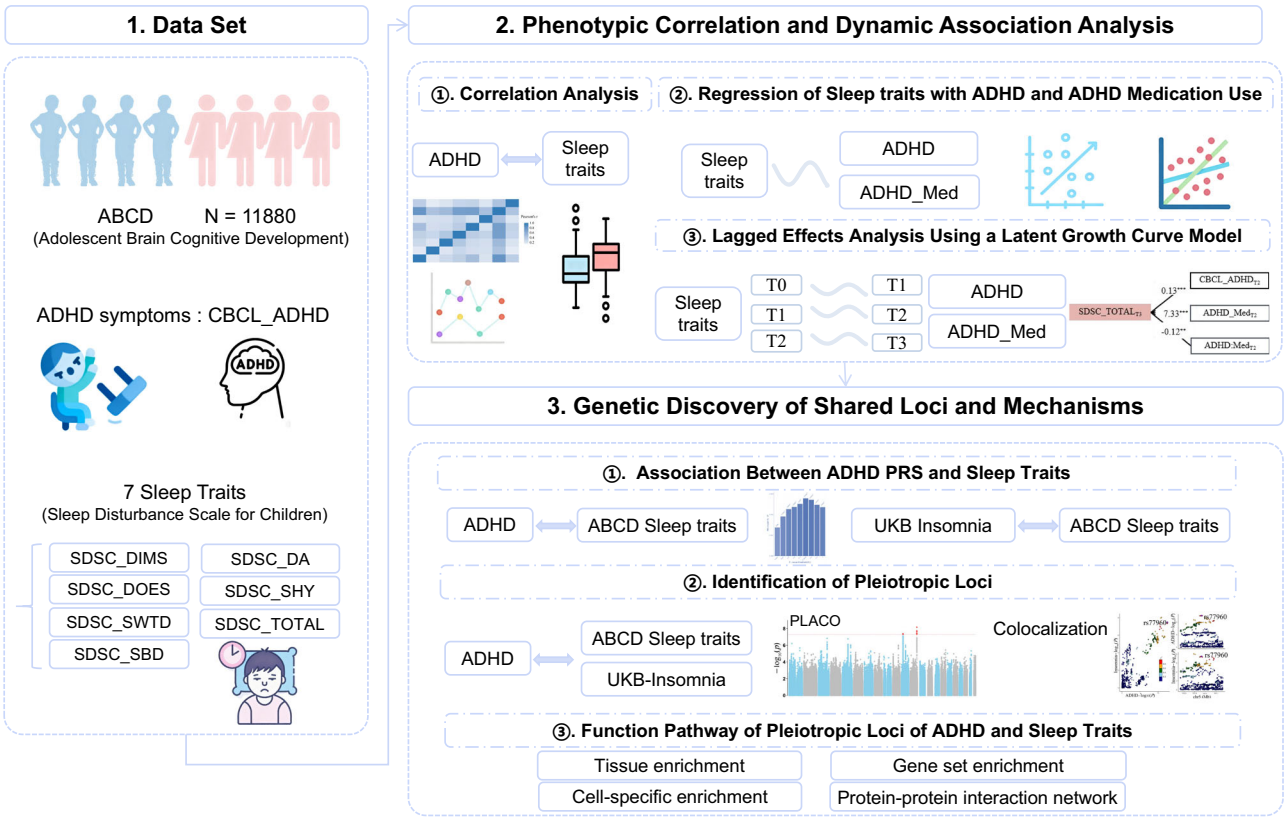



**Fig. 1 Study design.**

assessment and analysis of individual differences, which is crucial in developmental neuroscience. Specifically, in the context of ADHD and sleep problems, it is crucial to better understand how sleep problems evolve over time in individuals with ADHD and whether this trajectory is influenced by variables such as ADHD medications. By distinguishing between time-invariant (between-person) and time-varying (within-person) effects, Latent Growth Curve Modeling (LGCM) offers a robust and flexible analytical framework for testing an array of developmental hypotheses [20].

Twin studies have demonstrated that ADHD has a heritability of approximately 74% [21]. The most recent ADHD genome-wide association studies (GWAS) identified 27 genome-wide significant loci, and single nucleotide polymorphisms (SNPs) accounted for 14% of the heritability [22]. Previous research has indicated that the relationship between ADHD and sleep traits is shaped by both genetic and environmental factors, though the underlying mechanisms and causal relationships remain unclear [23]. Individuals with ADHD may be more susceptible to sleep disorders due to the combined influence of genetic predispositions. Twin studies [24] and ADHD GWAS have revealed a positive genetic correlation between ADHD and insomnia [25]. Additionally, Liu et al. found that insomnia is closely associated with the future development of ADHD [12]. A Mendelian Randomization study further supported a causal relationship, suggesting that insomnia, daytime napping, and short sleep duration contribute to ADHD [23]. Although the hypothesis proposed by Hvolby that shared neurobiological mechanisms contribute to both ADHD and sleep disturbances [26] is supported by evidence implicating the monoaminergic system in the pathophysiology of ADHD [27] and the regulation of sleep-wake cycles [28], the genetic underpinnings of this relationship remain poorly understood. A critical gap exists in the identification of pleiotropic SNPs or shared genetic risk factors mediating the ADHD-sleep association, precluding comorbid mechanistic insights and identify potential therapeutic targets. This underscores the need to characterize their genetic overlap and common pathways.

Hence, in this study, we aimed to explore the dynamic relationship of ADHD symptoms with several sleep traits using longitudinal data and model, and to analyze whether this correlation is influenced by factors such as ADHD medications use; then we further identify shared genetic pleiotropic loci between ADHD and sleep traits to try to explain the tight associations by investigating the biological functions of these pleiotropic genes (Fig. 1).

## MATERIALS AND METHODS

### Data sets

For the phenotype and dynamic correlation analyses, we utilized data from the Adolescent Brain Cognitive Development (ABCD) Study. The ABCD Study is an ongoing longitudinal and observational research initiative designed to explore brain development and child health. It includes children recruited from 21 sites across the United States (https://abcdstudy.org/study-sites/). The study began with a comprehensive baseline assessment (T0) of 11,880 children aged 9 to 10 years, followed by follow-up assessments at 1 to 3 years (T1, T2, T3). We included data from the ABCD Study data release 5.1.

### Measures

*ADHD symptoms and diagnosis of ADHD in ABCD.* The Achenbach's Child Behavior Checklist (CBCL) is a key tool for identifying problematic behaviors in children. In this analysis, we used T-scores from the ADHD DSM-5 Scale (CBCL_ADHD) as ADHD symptom data and a binary variable for ADHD diagnosis, based on parent reports from the Kiddie Schedule for Affective Disorders and Schizophrenia (KSAD-S). ADHD patients were defined as those meeting diagnostic criteria without comorbid conditions (Supplement Methods), while controls had neither ADHD diagnosis nor comorbid conditions. Notably, this binary ADHD diagnostic variable was only used to examine group differences in sleep problems between children with and without ADHD, whereas all other analyses employed the

**Table 1.** The characteristics of the variables used from Adolescent Brain Cognitive Development Study (n = 11288).

| **Variable** | **Value** | | | |
|---|---|---|---|---|
| Sex | Male 5887 (52.1%) / Female 5400 (47.9%) | | | |
| Race/ethnicity | White 5867 (51.9%) / Black 1695 (15%) / Hispanic 2292 (20.3%) / Asian 239 (2.1%) / Other 1193 (10.7%) | | | |
| **Phenotype** | **T0 (N/Mean/SD)** | **T1 (N/Mean/SD)** | **T2 (N/Mean/SD)** | **T3 (N/Mean/SD)** |
| Age | 11287/9.48/0.51 | 10639/10.48/0.64 | 10339/11.58/0.70 | 958/12.46/0.68 |
| BMI | 11283/18.76/4.09 | 10622/19.69/4.55 | 8701/20.68/4.88 | 3649/21.63/5.39 |
| CBCL_ADHD | 11281/53.23/5.64 | 10642/52.97/5.39 | 10333/53.09/5.31 | 9571/53.38/5.41 |
| ADHD_Diagnosis[a] | 770 (6.8%) | 845 (7.4%) | 544 (4.8%) | 496 (4.3%) |
| SDSC_DIMS | 11283/11.75/3.75 | 10640/11.94/3.78 | 10315/12.06/3.79 | 9548/12.22/3.95 |
| SDSC_DOES | 11284/6.95/2.44 | 10640/7.18/2.59 | 10315/7.14/2.58 | 9550/7.14/2.62 |
| SDSC_SWTD | 11259/8.19/2.63 | 10626/8.03/2.49 | 10314/7.79/2.42 | 9548/7.49/2.17 |
| SDSC_SBD | 11283/3.77/1.25 | 10640/3.73/1.19 | 10315/3.68/1.15 | 9548/3.59/1.06 |
| SDSC_SHY | 11283/2.44/1.18 | 10345/2.39/1.08 | 10028/2.34 /0.99 | 9280/2.29/0.89 |
| SDSC_DA | 11283/3.44/0.93 | 10345/3.35/0.78 | 10028/3.31/0.72 | 9281/3.25/0.66 |
| SDSC_TOTAL | 11258/36.54/8.25 | 10626/36.63/8.06 | 10314/36.32/8.07 | 9548/36.00/7.87 |
| Household income Mid[a] (≥50,000 and <100,000) | 2927 (25.9%) | 2702 (23.9%) | 2567 (22.7%) | 2328 (20.6%) |
| Household income High[a] (≥100,000) | 4319 (38.2%) | 4470 (39.5%) | 4581 (40.5%) | 4452 (39.4%) |
| Any ADHD medication[a] | 947 (8.3%) | 920 (8.1%) | 925 (8.1%) | _ |
| Any Sleep medication[a] | 24 (0.2%) | 21 (0.1%) | 19 (0.1%) | _ |
| Average Screen Time | 11263/3.79/3.06 | 10620/4.49/3.50 | _ | _ |

*BMI*, Body Mass Index; *CBCL*, Child Behavior Checklist; *CBCL_ADHD*, ADHD scale of CBCL; *SDSC*, Sleep Disturbance Scale for Children; *SDS_DIMS*, disorders of initiating and maintaining sleep (dims) from SDSC; *SDSC_DOES*, disorders of excessive somnolence (does) from SDSC; *SDSC_SWTD*, sleep–wake transition disorders (swtd) from SDSC; *SDSC_SBD*, sleep breathing disorders (sbd) from SDSC; *SDSC_SHY*, sleep hyperhidrosis (shy) from SDSC; *SDSC_DA*, disorders of arousal or nightmares (da) from SDSC; *SDSC_TOTAL*, total scores from SDSC; *Average Screen Time*, a weighted sum score to represent a daily STU, 5/7 x hours of STU in a typical weekday + 2/7 x hours of STU in a typical weekend day. *ADHD_Diagnosis*, diagnosed with ADHD. *T0*, baseline; *T1*, 1 year follow-up; *T2*, 2 year follow-up; *T3*, 3 year follow-up.
[a]The value for this variable is N (%).

continuous CBCL_ADHD T-scores. Detailed sample information is presented in Table 1.

*Sleep traits in ABCD*. We evaluated sleep traits in children using the Sleep Disturbance Scale for Children (SDSC), a 26-item parent-rated questionnaire based on the child's behavior over the preceding six months, which utilizes a 5-point Likert-type scale for responses. The SDSC comprises six subscales that assess different sleep-related issues [29], i) disorders of arousal or nightmares (SDSC_DA), ii) disorders of initiating and maintaining sleep (SDSC_DIMS), iii) disorders of excessive somnolence (SDSC_DOES), iv) sleep breathing disorders (SDSC_SBD), v) sleep hyperhidrosis (SDSC_SHY), and vi) sleep–wake transition disorders (SDSC_SWTD). Additionally, we included the total score (SDSC_TOTAL) as a measure of overall sleep disturbances. Higher scores indicate greater clinical severity of sleep disturbance. Details provided in the Table 1 and Supplementary Materials.

*Other relevant variables*. Household income was categorized into three levels (<$50 K, $50K-$100 K, >$100 K). Medication use for ADHD and sleep disorders was assessed via the PhenX Toolkit [30]. A weighted daily screen time score was derived from self-reported weekday and weekend usage (details provided in the Table S1 and Supplementary Materials).

*Genotype data and ADHD PRS calculation*. Genomic DNA from biospecimens in the ABCD database was genotyped using the Affymetrix NIDA Smoke Screen Array [31], with rigorous quality control applied to SNPs and individuals, utilizing PLINK for data processing (Supplement Methods). We computed polygenic risk score (PRS) of ADHD based on the recent ADHD GWAS [22] for each sleep trait using PRSice-2 [32]. The associations between the PRS and the target phenotypes were examined in linear regression models with age, gender, and genetic principal component (PC) 1–10 as covariates for each P-value threshold (Supplement Methods).

## Statistical analysis

*Phenotype correlation analysis*. We used Pearson's correlation to assess relationships between ADHD symptom scores and SDSC subscales/total scores at baseline and follow-up (1–3 years), with significance set at FDR < 0.05. T-tests was used to compare SDSC scores between ADHD and non-ADHD groups, using the same FDR threshold for significance. To examine the potential moderating role of ADHD medication in the relationship between ADHD symptoms and sleep problems, we performed multiple linear regression analyses. The models included sleep problem scores as the dependent variable and the interaction term between centered ADHD symptoms (to mitigate multicollinearity) and medication status (a binary variable) as the independent variable of interest, adjusting for age, sex, race/ethnicity, household income, screen use time, and sleep medication use. We validated the baseline findings using first-year follow-up data and conducted sensitivity analyses stratified by sex to assess the associations separately among children with and without ADHD medication. Prior to statistical testing, normality of distribution was examined using the Shapiro-Wilk test. Some variables showed right-skewed distributions. Given the large sample size of this study, parametric tests (Pearson correlation, independent samples t-test, and linear regression) were still employed based on the central limit theorem.

*Latent growth curve model analysis*. The latent growth curve model (LGCM) used in this study consisted of fixed and random effects to analyze sleep trait trajectories using the R package *lavaan* [33]. We evaluated model fit with RMSEA, SRMR, CFI, and TLI, aiming for good fit criteria. Sex, race and ADHD PRS were included as time-invariant covariates, while age, BMI, household income, sleep medication use, and ADHD symptoms, ADHD medication, and their interaction were served as time-varying covariates at four time points. This approach facilitated the disentanglement of between-person and within-person effects, allowing for a nuanced assessment of the lagged effects of factors such as ADHD symptoms and medication use on sleep. Detailed information is provided in Supplement Methods.

*Genetic correlation analysis.* We conducted PRS analysis to investigate the genetic correlation between ADHD and sleep traits. ADHD was used as base to get the best P-value threshold to predict the sleep traits (six subscales and a total score scale at baseline). Permutation n = 10000 was used to get empirical P-value. Empirical $P < 0.05$ was considered as significant.

*Identification of pleiotropic loci.* We performed GWAS on SDSC traits using PLINK, adjusting for age, sex, and 10 PCs. Pleiotropic loci between ADHD and sleep traits were identified using pleiotropic analysis under the composite null hypothesis (PLACO) [34]. This approach is based on the hypothesis that a genetic variant is associated with either none or only one of the examined traits. We utilized meta-analysis summary statistics for ADHD and three GWAS summary statistics datasets for SDSC to identify shared common SNPs. Genetic variants exceeding a predefined significance threshold (single-trait p-value > 10 − 4) for any trait were excluded to estimate the correlation coefficient (ρ) between Z-scores. Notably, PLACO demonstrates versatility by accommodating both independent traits and correlated summary statistics across studies, following Z-score decorrelation. To enhance power, we incorporated the GWAS summary data of insomnia data from UK Biobank (GWAS sample size N = 440,422), calculated its PRS associations with sleep traits in ABCD, and pleiotropic loci with ADHD using PLACO. Bayesian colocalization [35] (PP4 > 0.9) was used to validate the shared loci. Details are in Supplement Methods.

*Functional analysis of the pleiotropic loci.* To characterize the function of pleiotropic loci shared by ADHD and sleep from PLACO findings, we clumped SNPs within a ± 250 kb radius and linkage disequilibrium (LD) threshold of $r^2$>0.6 using FUMA [36]. Loci were annotated based on proximal genes, expression quantitative trait loci (eQTL), and chromatin interactions. Tissue enrichment analysis for the mapped genes was conducted with MAGMA, considering Bonferroni-corrected $P < 0.05$ as significant. Gene set enrichment was analyzed via Metascape with FDR < 0.05 for significance. Additionally, Web-based Cell-type Specific Enrichment Analysis of Genes (WebCSEA [37]) was used to assess the cell-type specificity across 1355 human tissue-cell types with significance set at Bonferroni-corrected P-value <0.05. Furthermore, we examined the enrichment of brain cell types by using single-cell transcriptomic data from three brain regions (Lake et al., 2018). Then, the protein-protein interaction network involved by the mapped genes were obtained from String (confidence score >0.4) and the hub genes were identified from this network using the Maximal Clique Centrality (MCC) algorithm.

## RESULTS

### Phenotype correlation and effect by ADHD medication

Our analysis firstly verified the fundamental phenotypic associations between ADHD symptoms and sleep traits. We observed significant positive correlations between CBCL_ADHD scores and both the total sleep score and subscales at baseline (Fig. 2A), which were consistently maintained throughout the follow-up period (Fig. 2B). This initial analysis served not only as a quality control measure, by replicating well-established effects, but also streamlined our subsequent analysis by identifying four key sleep subscales for further investigation. To quantify the differences in specific sleep traits between individuals with and without ADHD after excluding comorbidities, we conducted between-group comparisons on the four key sleep subscales (SDSC_DIMS, SDSC_DOES, SDSC_SWTD, and SDSC_TOTAL). The results confirmed that the ADHD group scored significantly higher on all these measures, with Cohen's d values indicating strong effects (Figure S1A-D).

To delineate the association patterns between sleep characteristics and ADHD-related variables, we constructed multiple linear

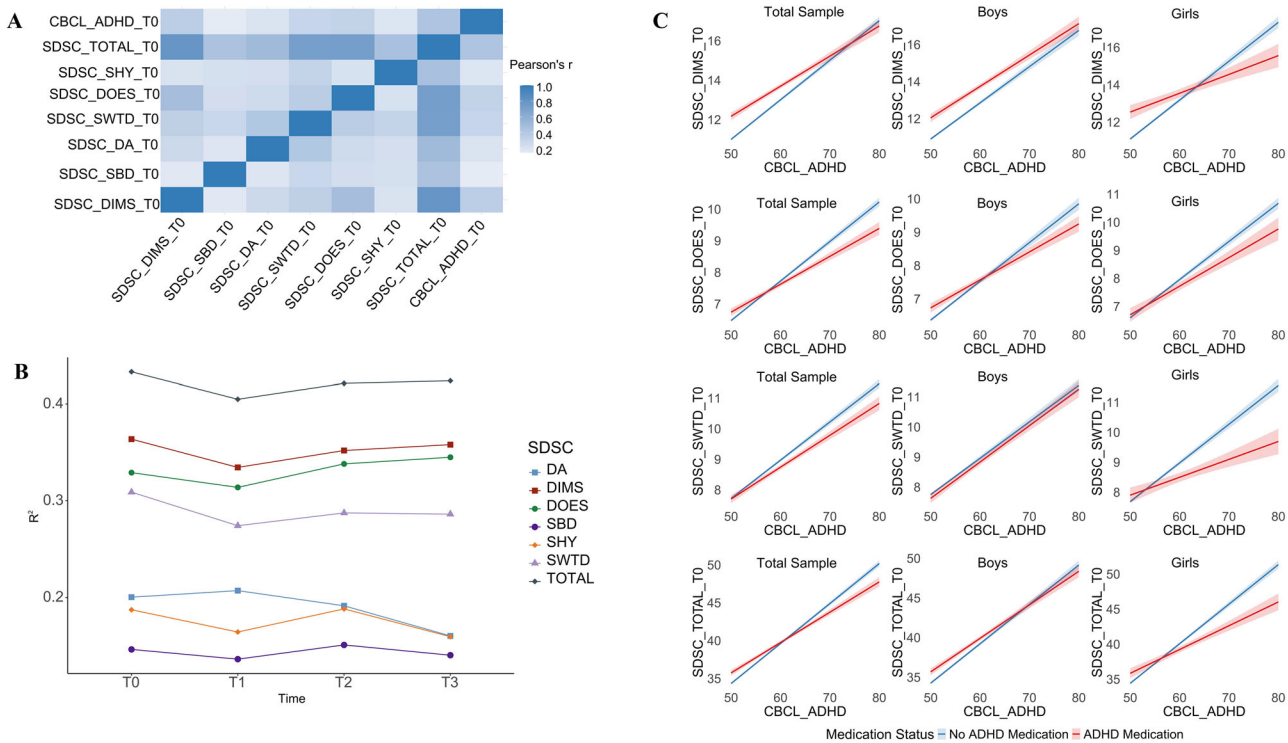



**Fig. 2 Phenotype analysis and association of ADHD medications with ADHD symptoms and sleep traits. A** The correlation of ADHD symptom score (CBCL_ADHD) with the SDSC subscale at baseline. All of the correlation were significant. **B** line plots of correlations between ADHD symptom (CBCL_ADHD) and sleep scales at baseline and 1-3 follow-up years. **C** Multivariate regression models examining the ADHD-sleep relationship. Factor: Medication use (Yes/No); Conducted in: Total sample, Boys, and Girls separately. The red squares indicate a significant association of ADHD symptoms with the sleep phenotype in the model, while the solid blue circles represent a significant interaction term between ADHD symptoms and medication use. Shaded areas represent confidence intervals. All significant findings have been corrected for multiple comparisons using the False discovery rate (FDR) method. T0: baseline; SDSC_DIMS: disorders of initiating and maintaining sleep (dims) from SDSC; SDSC_DOES: disorders of excessive somnolence (does) from SDSC; SDSC_SWTD: sleep-wake transition disorders (swtd) from SDSC; SDSC_TOTAL: total scores from SDSC; CBCL_ADHD: Child Behavior Checklist ADHD scale.

**Table 2.** Association of Sleep Traits with ADHD Symptoms and Medication Use at Baseline Adolescent Brain Cognitive Development Study.

| | SDSC_DIMS ~ CBCL_ADHD | | | SDSC_DOES ~ CBCL_ADHD | | | SDSC_SWTD ~ CBCL_ADHD | | | SDSC_TOTAL ~ CBCL_ADHD | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Variable** | **B** | **SE** | **FDR** | **B** | **SE** | **FDR** | **B** | **SE** | **FDR** | **B** | **SE** | **FDR** |
| CBCL_ADHD [a] | 0.20 | 0.01 | $1.94 \times 10^{-157}$ | 0.13 | 0.004 | $3.19 \times 10^{-55}$ | 0.12 | 0.01 | $1.65 \times 10^{-128}$ | 0.53 | 0.01 | $3.22 \times 10^{-284}$ |
| Any ADHD medication | 1.08 | 0.16 | $6.29 \times 10^{-11}$ | 0.21 | 0.10 | 0.07 | −0.10 | 0.11 | 0.47 | 1.08 | 0.30 | $1.27 \times 10^{-3}$ |
| Age | 0.12 | 0.06 | 0.07 | 0.09 | 0.04 | 0.03 | −0.02 | 0.04 | 0.69 | 0.13 | 0.12 | 0.37 |
| Sex2 | 0.23 | 0.06 | $6.53 \times 10^{-4}$ | 0.21 | 0.04 | $8.26 \times 10^{-7}$ | −0.08 | 0.04 | 0.11 | 0.19 | 0.12 | 0.17 |
| Race2 | 0.02 | 0.10 | 0.86 | −0.14 | 0.07 | 0.07 | −0.61 | 0.07 | $2.40 \times 10^{-16}$ | −0.59 | 0.20 | $7.45 \times 10^{-3}$ |
| Race3 | −0.55 | 0.09 | $1.76 \times 10^{-9}$ | −0.15 | 0.05 | 0.01 | −0.05 | 0.06 | 0.48 | −0.77 | 0.17 | $2.15 \times 10^{-5}$ |
| Race4 | −0.25 | 0.21 | 0.33 | −0.03 | 0.13 | 0.86 | 0.09 | 0.15 | 0.65 | 0.12 | 0.41 | 0.82 |
| Race5 | 0.09 | 0.11 | 0.48 | 0.004 | 0.07 | 0.97 | 0.16 | 0.07 | 0.05 | 0.48 | 0.21 | 0.04 |
| Household income Mid (≥ 50,000 and <100,000) | 0.12 | 0.08 | 0.21 | −0.02 | 0.05 | 0.69 | 0.09 | 0.06 | 0.19 | 0.078 | 0.16 | 0.70 |
| Household income High (≥ 100,000) | −0.18 | 0.08 | 0.05 | −0.17 | 0.05 | $1.91 \times 10^{-3}$ | $1.19 \times 10^{-3}$ | 0.06 | 0.98 | −0.57 | 0.16 | 0.001 |
| BMI | 0.02 | 0.01 | 0.05 | 0.01 | 0.004 | $4.87 \times 10^{-3}$ | 0.01 | 0.01 | 0.08 | 0.09 | 0.02 | $6.09 \times 10^{-9}$ |
| Average Screen Time [b] | 0.05 | 0.01 | $1.42 \times 10^{-5}$ | 0.01 | 0.009 | 0.07 | −0.01 | 0.01 | 0.59 | 0.06 | 0.02 | 0.01 |
| Any Sleep medication | 1.26 | 0.82 | 0.18 | 0.24 | 0.51 | 0.70 | 0.72 | 0.56 | 0.28 | 2.84 | 1.58 | 0.11 |
| CBCL_ADHD * Any ADHD medication [c] | −0.05 | 0.02 | 0.01 | −0.04 | 0.01 | $9.96 \times 10^{-4}$ | −0.02 | 0.01 | 0.12 | −0.13 | 0.03 | $2.29 \times 10^{-4}$ |

Sex2: A binary variable indicating the participant's sex (1 = Female, 0 = Male). Race2-5: Race/ethnicity was coded as a series of binary variables (1 = yes, 0 = no) using White as the reference category: Race2: Black, Race3: Hispanic, Race4: Asian, Race5: Other.
[a]The Mean-centered CBCL_ADHD scale score. A score of 0 represents the sample mean, and units represent deviations from this mean.
[b]A weighted sum score to represent a daily STU,5/7 x hours of STU in a typical weekday + 2/7 x hours of STU in a typical weekend day.
[c]Interaction term was introduced in the model.

regression models. The analysis revealed significant associations between ADHD symptoms and all four key sleep traits (SDSC_DIMS, SDSC_DOES, SDSC_SWTD, SDSC_TOTAL, Table 2, Fig. 2C). Notably, the associations for DIMS, DOES, and TOTAL sleep traits were significantly moderated by an interaction between ADHD symptoms and medication use (SDSC_DIMS: $\beta = -0.05$, FDR $q = 0.007$; SDSC_DOES: $\beta = -0.04$, FDR $q = 9.97 \times 10^{-4}$; SDSC_TOTAL: $\beta = -0.13$, FDR $q = 2.29 \times 10^{-4}$), indicating that the association between sleep problems and ADHD symptom severity was attenuated in medication users. These findings, including the moderating role of medication, remained consistent in one-year follow-up data (SDSC_DIMS: $\beta = -0.05$, FDR $q = 2.59 \times 10^{-3}$; SDSC_DOES: $\beta = -0.03$, FDR $q = 0.01$; SDSC_TOTAL: $\beta = -0.09$, FDR $q = 2.30 \times 10^{-3}$, Table S2, Fig. 2C), demonstrating their robustness and temporal stability (Table S2).

Sex-stratified analyses revealed that the main associations between sleep phenotypes and ADHD symptoms were significant in both boys and girls. However, the moderating effect of medication use differed by sex. In boys, a significant interaction between ADHD symptoms and medication was observed only for the SDSC_DOES subscale (SDSC_DOES: $\beta = -0.03$, FDR $q = 0.03$, Table S2, Fig. 2C). In contrast, among girls, this moderating effect was significant across three phenotypes: SDSC_DIMS, SDSC_SWTD, and SDSC_TOTAL (SDSC_DIMS: $\beta = -0.11$, FDR $q = 2.32 \times 10^{-3}$; SDSC_SWTD: $\beta = -0.07$, FDR $q = 3.60 \times 10^{-3}$; SDSC_TOTAL: $\beta = -0.22$, FDR $q = 7.31 \times 10^{-4}$, Table S2, Fig. 2C). However, it should be noted that both ADHD symptoms and sleep characteristics were assessed using parent-reported instruments, which carries a risk of reporting bias.

### Longitudinal Dynamic relationship of ADHD and sleep traits

To investigate whether the moderating effect of interaction on the relationship between sleep traits and ADHD exerts an impact on the subsequent year (whether a lag effect exists), we examined their dynamic relationship using latent growth curve model. We incorporated sex, race and ADHD PRS as time-invariant covariates (TIVs) in the LGCM to evaluate their influence on ADHD symptoms and sleep traits, thereby assessing between-person differences. Model fit indices were satisfactory (Table S3). Higher ADHD PRS significantly predicted the initial levels (intercept: *i*) of SDSC_DIMS, SDSC_DOES, and SDSC_TOTAL, suggesting increased sleep difficulties in individuals with greater ADHD genetic risk (Table S4).

ADHD symptoms, medication use, and their interaction term were included as time-varying covariates (TVCs) in the model to investigate their lagged effects on sleep characteristics at the within-person level. The analysis revealed that CBCL_ADHD scores from T0 to T2 significantly predicted the four sleep traits at timepoints T1 to T3 (Fig. 3A–D, Table S4), indicating that prior ADHD symptom levels have a stable predictive value for subsequent sleep outcomes. Furthermore, the interaction term exhibited a significant lagged effect for the SDSC_DIMS and SDSC_TOTAL subscales, suggesting that ADHD medication weakens the association between ADHD symptoms and sleep problems in the subsequent year.

### Genetic correlation of ADHD and sleep traits

At the basis of phenotype correlation, we further investigated the effects of PRS for ADHD on the risk of three sleep subscales and sleep total scale (i.e. SDSC_DIMS, SDSC_DOES, SDSC_SWTD, SDSC_TOTAL) from the ABCD database. ADHD PRS showed statistically significant associations with SDSC_DIMS, SDSC_DOES, SDSC_TOTAL (Figure S2A–C). Notably, the proportion of variance explained by ADHD PRS was very modest, ranging from 0.28% to 0.30% (ADHD with SDSC_DIMS: maximum at P-value Threshold = 0.39, $R^2 = 0.28\%$, Empirical-P $= 3.0 \times 10^{-4}$; ADHD with SDSC_DOES: maximum at P-value Threshold = 0.46, $R^2 = 0.28\%$, Empirical-P $= 1.0 \times 10^{-5}$, ADHD with SDSC_TOTAL: maximum at P-value Threshold = 0.39, $R^2 = 0.30\%$, Empirical-P $= 1.0 \times 10^{-4}$).

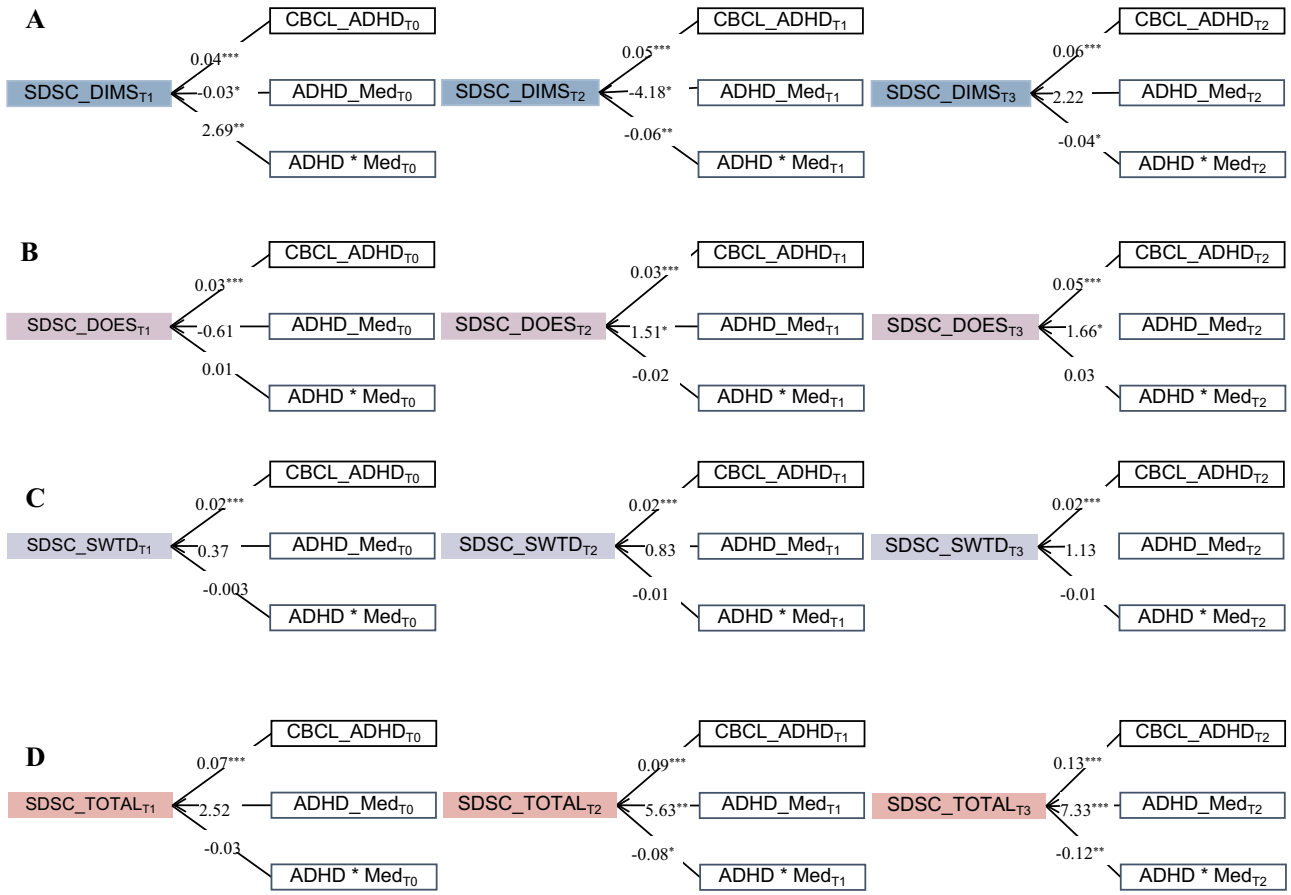



**Fig. 3 The result of latent growth curve model (LGCM) of sleep phenotypes predicted by time-invariant covariates (TIC) and lagged time-varying covariates (TVC). A** SDSC_DIMS; **B** SDSC_DOES; **C**: SDSC_SWTD; **D** SDSC_TOTAL. All models included ADHD PRS, sex, race as TIC, age, BMI, household income, sleep medication use, ADHD symptoms, ADHD medication and their interaction term measured at T0, T1, and T2 as lagged TVCs. T0: baseline; T1:1 year follow-up; T2: 2 year follow-up. * indicates *P*-value < 0.05, ** indicates *P*-value < 0.01, *** indicates *P*-value < 0.001.

Although these associations reached statistical significance, their small effect sizes indicate limited practical explanatory power at the individual level. No significant correlation was found with SDSC_SWTD (Empirical-P = 0.27) (Figure S2D).

### Pleiotropic loci of ADHD and sleep traits

We investigated the pleiotropic loci between ADHD and sleep traits (SDSC_DIMS, SDSC_DOES, SDSC_TOTAL) based on genetic correlations identified in the PRS analysis. Firstly, GWAS revealed 2, 4, and 3 significant loci for SDSC_DIMS, SDSC_DOES, and SDSC_TOTAL, respectively, mapping to 64 genes (Figure S3A-C, Table S5). Then, PLACO analysis of the GWAS summary data showed 0, 2, 0 significant loci for ADHD and SDSC_DIMS, SDSC_DOES, SDSC_TOTAL, respectively ($P_{PLACO} < 5 \times 10^{-8}$). The lead SNPs for the pleiotropic loci were rs7942482 and rs1790123. Notably, rs1790123, a key pleiotropic locus on chromosome 12, mapped to 30 genes by positional, eQTL or chromatin interaction (Table 3), including 7 genes mapped by cis-eQTL SNPs who influence gene expression levels, 27 genes mapped by 3D DNA–DNA interactions, and 7 genes were annotated by both eQTL mapping and chromatin interactions (Fig. 4C), though it did not reach significance in the original GWAS ($P > 5 \times 10^{-8}$).

Due to the small sample size of the ABCD database, we further used the GWAS summary data of insomnia from UK Biobank for PLACO analysis. Given inherent differences in age, phenotype definition, and cohort design between pediatric sleep phenotypes in ABCD and adult insomnia GWAS data from UK Biobank, the following results should be regarded as exploratory. Firstly, PRS analysis showed significant positive correlations between UKB insomnia PRS and SDSC_DIMS/SDSC_TOTAL (Figure S2E-F), while LD Score regression revealed a strong genetic correlation between ADHD and insomnia ($P < 4.8 \times 10^{-37}$; $r_g = 0.32$, s.e.=0.03).

Then, we explored shared pleiotropic loci between insomnia and ADHD using PLACO, identifying 14 lead SNPs across 12 genomic risk loci (Fig. 4B, Table 3). Among these, two loci were significantly associated with ADHD and insomnia in the original GWAS, two with only ADHD, three with only insomnia, and seven with neither. These loci mapped to 98 genes through positional, eQTL or chromatin interaction analyses (Table 3), including 27 cis-eQTL SNPs, 74 chromatin interaction genes, and 14 genes from both methods (Fig. 4D, Figure S5).

Bayesian colocalization tests were conducted on the two ADHD and SDSC_DOES pleiotropic loci and fourteen ADHD and insomnia pleiotropic loci (Table 3) from PLACO results. One shared locus for ADHD and SDSC_DOES (Fig. 4E and Table 3) (PP4 > 90%) and four loci for ADHD and insomnia (Fig. 4F–I and Table 3) (PP4 > 90%) indicated strong causal relationships. The lead SNP rs1790123 ($PPLACO < 6.89 \times 10^{-9}$) is located in the intron of MPHOSPH9, while rs77960 ($PPLACO < 6.54 \times 10^{-15}$, PP4 = 98.8%) is located in the intron of lincRNA RP11-6N13.1. SNP rs62264764 ($PPLACO < 1.01 \times 10^{-9}$, PP4 = 91%) was mapped to gene LSAMP:RP11-384F7.2. The other SNP rs1476535 ($PPLACO < 5.69 \times 10^{-13}$, PP4 = 90%) was mapped to gene FOXP2, and SNP rs10815977 ($PPLACO < 1.73 \times 10^{-8}$, PP4 92.8%) was nearby gene PTPRD:RP11-75C9.1.

**Table 3.** The pleiotropic loci identified by PLACO for ADHD and sleep traits, and the *coloc* colocalization posterior probability (PP4) for the lead SNPs for each locus.

| ADHD and sleep traits | Genomic Locus | Lead SNP | Chr | Pos | Lead SNP from PLACO analysis | | | | | *coloc* analysis | | Mapped genes [a] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | $P_{PLACO}$ | $\beta_{ADHD}$ | $P_{ADHD}$ | $\beta_{sleep}$ | $P_{sleep}$ | PP3 | PP4 | |
| PLACO for ADHD and SDSC_DOES | 1 | rs7942482 | 11 | 28677264 | $4.24 \times 10^{-8}$ | 1.0739 | $1.96 \times 10^{-13}$ | −0.1002 | 0.0102 | 0.04 | 0.10 | RP11-960D24.1 |
| | 2 | rs1790123 | 12 | 123659542 | $6.89 \times 10^{-9}$ | 0.94952 | $7.78 \times 10^{-6}$ | 0.2098 | $9.92 \times 10^{-6}$ | 0.03 | 0.90 | MPHOSPH9, ZCCHC8, KDM2B, PITPNM2, RSRC2, OGFOD2, CDK2AP1, GTF2H3, EIF2B1, DDX55, CCDC92, CLIP1, CCDC62, HIP1R, C12orf65, SBNO1, VPS33A, DENR, ABCB9, RILPL2, TCTN2, MLXIP, FAM101A, ZNF664, ARL6IP4, ORAI1, SETD8, KNTC1, ATP6V0A2, RP11-512M8.5 |
| PLACO for ADHD and Insomnia | 1 | rs11579563 | 1 | 87721876 | $7.39\times 10^{-9}$ | 1.0374 | $2.19 \times 10^{-4}$ | −0.0089 | $1.95 \times 10^{-7}$ | 0.05 | 0.73 | LMO4 |
| | 2 | rs4671328 | 2 | 58935282 | $2.74\times 10^{-8}$ | 1.0311 | $4.48 \times 10^{-3}$ | −0.0102 | $6.39 \times 10^{-10}$ | 0.22 | 0.01 | LINC01122, BCL11A |
| | 3 | rs11693221 | 2 | 66799986 | $2.64\times 10^{-11}$ | 0.9328 | $2.81 \times 10^{-3}$ | −0.0448 | $1.45 \times 10^{-27}$ | 0.02 | 0.17 | MEIS1, RAB1A, ACTR2 |
| | 4 | rs11130182 | 3 | 49206767 | $7.48\times 10^{-9}$ | 1.0392 | $9.27 \times 10^{-5}$ | 0.0100 | $5.28 \times 10^{-9}$ | 0.07 | 0.82 | KLHDC8B |
| | 4 | rs9843653 | 3 | 49920571 | $2.53\times 10^{-9}$ | 0.9573 | $3.51 \times 10^{-6}$ | −0.0084 | $2.07 \times 10^{-7}$ | 0.87 | 0.11 | MST1R, SEMA3F, RBM5, RBM6, ZMYND10, CACNA2D2, CELSR3, UQCRC1, FAM120A, RHOA, HYAL2, RASSF1, IP6K2, PRKAR2A, USP4, GNAT1, GNAT2, TUSC2, NPRL2, CYB561D2, TMEM115, AMT, TCTA, NICN1, UCN2, CDC25A, CAMP, ZNF589, PLXNB1, CCDC51, ATRIP, BSN, APEH, RNF123, CAMKV, MON1A, LAMB2, USP19, QARS, NME6, DAG1, CCDC36, MST1, GMPPB, AMIGO3, IP6K1, CCDC71, ARIH2, IMPDH2, NDUFAF3, DALRD3, WDR6, P4HTM, SLC25A20, LSMEM2,UBA7, TRAIP, LSAMP, FAM212A, KLHDC8B, HYAL3, CDHR4, C3orf62, QRICH1, NCKIPSD, TREX1, ARIH2OS, RP11-3B7.1, CTD-2330K9.3, SLC26A6, TMA7, GPX1, C3orf84, NAT6, XXcos-LUCA11.5, PTPN23, SLC26A6, GNAI2, CTD-2330K9.3 |
| | 5 | rs62264764 | 3 | 117639575 | $1.01\times 10^{-9}$ | 1.0668 | $4.52 \times 10^{-7}$ | 0.0114 | $6.00 \times 10^{-7}$ | 0.08 | 0.91 | LSAMP:RP11-384F7.2 |
| | 6 | rs28657372 | 4 | 105375274 | $8.36\times 10^{-9}$ | 1.0373 | $1.93 \times 10^{-4}$ | −0.0087 | $2.86 \times 10^{-7}$ | 0.05 | 0.17 | CXXC4, PPA2, GSTCD, |
| | 7 | rs6452785 | 5 | 87685500 | $6.42\times 10^{-16}$ | 1.0574 | $2.00 \times 10^{-9}$ | −0.0010 | $8.08 \times 10^{-10}$ | 1 | 0.00 | TMEM161B-AS1, POLR3G, CCNH, RASA1, CETN3, TMEM161B, GPR98, LYSMD3, MBLAC2, |

**Table 3.** continued

| ADHD and sleep traits | Genomic Locus | Lead SNP | Chr | Pos | Lead SNP from PLACO analysis | | | | | *coloc* analysis | | Mapped genes [a] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | $P_{PLACO}$ | $\beta_{ADHD}$ | $P_{ADHD}$ | $\beta_{sleep}$ | $P_{sleep}$ | PP3 | PP4 | |
| | 7 | rs681446 | 5 | 88054292 | $2.78 \times 10^{-9}$ | 0.9589 | $1.13 \times 10^{-5}$ | 0.0077 | $3.04 \times 10^{-6}$ | 0.34 | 0.59 | MEF2C |
| | 8 | rs77960 | 5 | 103964585 | $6.54 \times 10^{-15}$ | 0.9294 | $2.46 \times 10^{-13}$ | −0.0095 | $3.34 \times 10^{-8}$ | 0.01 | 0.99 | RP11-6N13.1 |
| | 9 | rs1476535 | 7 | 114071035 | $5.69 \times 10^{-13}$ | 0.9324 | $2.03 \times 10^{-10}$ | −0.0090 | $5.04 \times 10^{-8}$ | 0.09 | 0.90 | FOXP2, MDFIC, |
| | 10 | rs10815977 | 9 | 8860355 | $1.73 \times 10^{-8}$ | 0.9590 | $2.07 \times 10^{-5}$ | −0.0083 | $3.90 \times 10^{-7}$ | 0 | 0.93 | PTPRD |
| | 11 | rs10821168 | 9 | 96358301 | $1.56 \times 10^{-8}$ | 1.0439 | $8.30 \times 10^{-6}$ | 0.0084 | $8.79 \times 10^{-7}$ | 0.16 | 0.62 | PHF2, HIATL1, FAM120AOS, FAM120A |
| | 12 | rs12577364 | 11 | 28658030 | $8.55 \times 10^{-9}$ | 1.0742 | $1.60 \times 10^{-13}$ | 0.0053 | 0.0016 | 0.02 | 0.06 | RP11-960D24.1 |

A high PP4 for a SNP indicates high probability of being the common causal SNP for both ADHD and sleep traits.
[a]mapped genes were based on positional, eQTL mappings or chromatin interaction from FUMA. Positional mapping with maximum distance 10 kb is performed. In eQTL mapping, BRAINEAC and GTEx v8 brain were selected as tissue types, eQTL P-value threshold used FDR <0.05.

### Function of pleiotropic loci of ADHD and sleep traits

Following pleiotropic locus analysis, exploratory tissue enrichment analysis using MAGMA revealed that ADHD and SDSC_DOES PLACO results were enriched in the pituitary (Figure S6A). These observations align with previous reports that dysregulation of the hypothalamic-pituitary-adrenocortical axis and circadian rhythms may be relevant to sleep regulation and attention [38]. GO enrichment analysis showed significant associations with translation factor activity, negative regulation of organelle, centrosome, transcription coregulator activityand mitotic cell cycle (Figure S6B). Enriched tissue-cell types included myeloid and erythroid progenitors, hematopoietic stem cells, and plasma cells, all crucial for immune function and linked to sleep [39] and ADHD [40] (Figure S6C).

As an exploratory extension, pleiotropic loci analysis for ADHD and insomnia revealed enrichment in brain-related tissues, particularly the brain cortex, frontal cortex (BA9), and anterior cingulate cortex (BA24) (Fig. 5A). GO enrichment analysis highlighted pathways potentially relevant to synapse organization and regulation of growth (Fig. 5B). The 98 mapped genes were significantly enriched in Lake_2017_VisualCortex_Ex5a, consistent with a potential involvement of excitatory neurons in the visual cortex, a region previously associated with for ADHD [41] and sleep [42] (Figure S6D).

To explore potential functional themes among the 128 genes shared between ADHD and sleep problems (SDSC_DOES and insomnia), we performed exploratory analyses using brain transcriptome data to define their enrichment patterns across diverse neural cell types [43]. The results revealed a significant enrichment of these genes in eight cell types in the prefrontal cortex (six excitatory and two inhibitory neurons), seventeen in the visual cortex (eleven excitatory and six inhibitory neurons), and one microglial cell (Mic) in the cerebellum (Fig. 5C). Furthermore, we constructed a protein-protein interaction (PPI) network for the 128 shared genes. Genes with literature evidence of association with neurodevelopmental diseases or sleep related traits were marked with different colors (Table S6). Using the Maximal Clique Centrality (MCC) algorithm, we identified 10 hub genes. Functional enrichment analysis indicated that these 10 hub genes are involved in growth-related pathways, specifically regulating cell growth (Fig. 5D).

## DISCUSSION

At the descriptive phenotypic level, this study revealed a significant positive correlation between ADHD symptom scores and sleep scale scores over time, particularly for SDSC_DIMS, SDSC_DOES, SDSC_SWTD, and SDSC_TOTAL. Children with higher ADHD symptom scores associated with more severe difficulties in sleep initiation, maintenance, excessive daytime sleepiness, and sleep-wake transitions. These findings are consistent with prior research demonstrating that ADHD, as one of the most extensively studied neurodevelopmental disorders (NDDs), is frequently associated with significant alterations in sleep patterns, including difficulties in falling asleep, impaired sleep maintenance, frequent nocturnal arousals, and daytime lethargy [44, 45].

Building on these descriptive phenotypic observations, we further conducted sex-stratified multiple linear regression analyses to examine the relationship between ADHD and sleep, particularly the moderating role of medication. As expected, ADHD symptoms consistently demonstrated a robust positive association with sleep problems. Interestingly, we found a significant negative interaction between ADHD symptoms and medication use, indicating that medication attenuated the association between ADHD symptoms and sleep. This indicates that although the medication itself may have some side effects on sleep (as shown by its significant β), it moderates the association between ADHD core symptoms and sleep problems. This pattern of moderation was

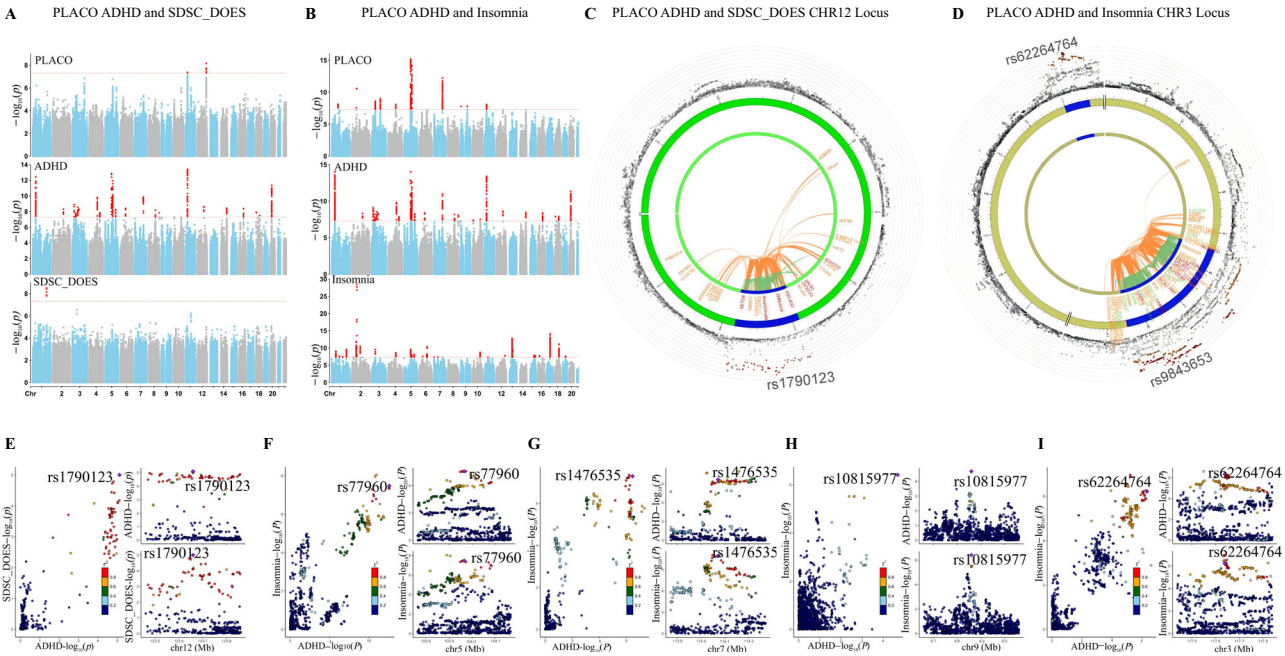



**Fig. 4 The results of PLACO, chromatin interaction and *COLOC* for ADHD and sleep traits. A–B** GWAS Manhattan plots for PLACO, ADHD, and sleep traits from top to bottom. The top represents the Manhattan plot of PLACO P-values for the pleiotropic association of ADHD with sleep traits. The red dashed line represents the threshold for genome-wide significant association ($P = 5 \times 10^{-8}$). Red dots represent genome-wide significant loci. **A** the results of PLACO for ADHD and SDSC_DOES. **B** the results of PLACO for ADHD and Insomnia. **C–D** chromatin interaction mapping circos plots for the pleiotropic sites of ADHD and DSDC_DOES (**C**) or insomnia (**D**). If the gene is mapped only by chromatin interactions or only by eQTLs, it is colored orange or green, respectively. When the gene is mapped by both, it is colored red. Links colored orange are chromatin interactions. Links colored green are eQTLs. **E** one locus that showed strong single-trait association signals in colocalization analyses (PP4 > 90%) for ADHD and SDSC_DOES. **F–I** four loci that all showed strong single-trait association signals in colocalization analyses (PP4 > 90%) for ADHD and Insomnia.

more pronounced in girls (Fig. 2), despite literature suggesting that girls with ADHD are less likely to receive pharmacological treatment [46, 47]. This finding may potentially stem from the instability inherent in the identified subgroups. Of note, the outcome measures in the present study are not objective sleep phenotypes, nor do they represent clinically defined sleep disorders.

Building on these longitudinal lagged association analyses, we used LGCM to investigate the temporal dynamics of ADHD symptoms on sleep traits. A key strength of this approach is its ability to distinguish between-person effects (time-invariant factors) and within-person effects (time-varying factors), which allowed us to examine sex differences, genetic influences, and associations with ADHD medication use in a manner not commonly applied in previous research. In the model incorporating time-invariant covariates (i.e. sex, PRS_ADHD), significant sex differences were observed at the between-person level. Girls showed significantly higher initial levels (intercepts) of disorders of initiating and maintaining sleep (SDSC_DIMS, Table S2) and disorders of excessive somnolence (SDSC_DOES, Table S2) than boys. Consistent associations were also detected between ADHD polygenic risk score (PRS) and baseline intercepts for SDSC_DIMS and SDSC_DOES. Therefore, we hypothesize that genetic susceptibility to ADHD may influence an individual's baseline level of these specific sleep problems. Regarding within-person effects, consistent and significant lagged associations were observed across time points, such that ADHD symptoms, ADHD medication use, and their interaction on subsequent sleep problems. These longitudinal associations support temporally ordered relationships but not causal effects.

Building upon exploratory post-GWAS analyses, we further investigated pleiotropic loci to generate hypotheses regarding the potential shared biological mechanisms underlying ADHD and sleep traits. Our analysis identified 30 genes associated with pleiotropic loci linking ADHD and SDSC_DOES, along with 98 genes associated with ADHD and insomnia. Enrichment analysis across neural cell types (Fig. 5C) revealed diverse excitatory neuron subtypes in both the visual cortex (VC) and frontal cortex (FC), including deep-layer subtypes Ex5a, Ex5b, Ex6a, and Ex6b. Notably, these deep-layer excitatory neurons are critical for long-range corticocortical and corticofugal projections [48] involved in executive control and arousal regulation, core neurobiological domains disrupted in both ADHD and sleep disturbances. Notably, we identified VC-specific subtypes Ex3b, Ex3c, and Ex3d in the VC, corroborating previous findings [43]. These VC excitatory neurons highly expressed markers such as SLC17A7 [49], which are associated with synaptic signaling and visual perception. Within established frameworks of circadian and visual pathway regulation of sleep, the visual cortex serves as a key relay for photic input to the suprachiasmatic nucleus (SCN) [50], and aberrant excitatory signaling in this region has been proposed to disrupt circadian entrainment and sleep–wake timing frequently observed in ADHD [51]. Furthermore, NREM sleep modulates neural synchrony in the visual cortex, thereby influencing cognitive performance [52]. Our analysis revealed significant enrichment of inhibitory neurons in the frontal cortex and visual cortex. Notably, the In6 subtype was prominently enriched in the visual cortex. The In6a subpopulation specifically encircles layer 4 [43], which is the primary target of thalamic visual inputs and, as noted earlier, is functionally implicated in synaptic neurotransmission and sleep [53]. Enrichment of inhibitory neuron subtypes such as In6a points to disrupted regional E/I balance as a shared cellular substrate for both ADHD pathophysiology and sleep dysregulation [51]. Prefrontal and visual cortical E/I imbalance is known to impair top-down inhibitory control and promote cortical hyperarousal, both of which contribute to attentional deficits and difficulties initiating and maintaining sleep [54].

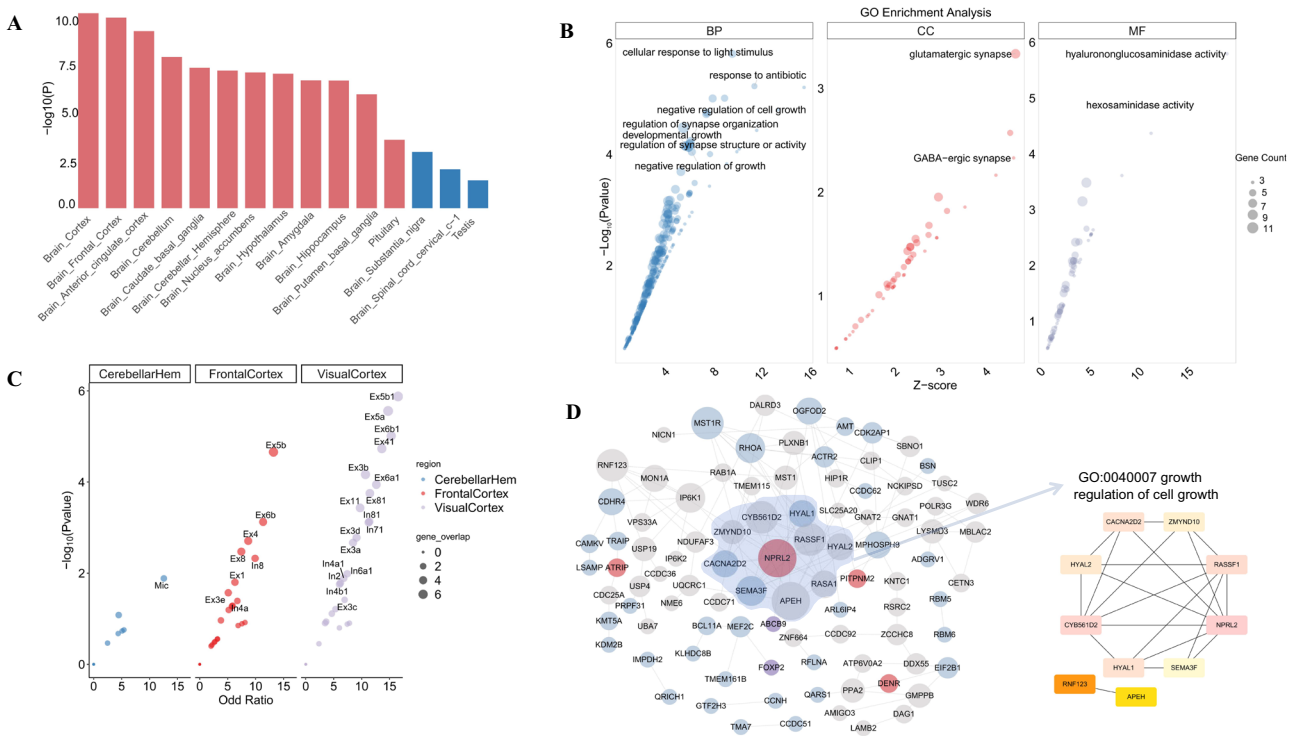



**Fig. 5 Functional analysis for the pleiotropic loci genes of ADHD and sleep traits. A** tissue enrichment analysis results of PLACO result of ADHD and insomnia using MAGMA. Genome-wide significance was set at 0.05 / (the number of tested tissues). **B** Go enrichment analysis of ADHD and insomnia pleiotropic loci genes in Metascape. **C** leveraging single-cell transcriptomic data from three brain regions, we performed enrichment analysis for a set of 128 genes across distinct cell types [43]. The names of cell types with P-value <0.05 were shown. **D** gene interaction network for pleiotropic loci of ADHD and sleep traits. The protein-protein interactions were from STRING database at confidence score = 0.4. This plot was generated by Cytoscape, nodes denoted for genes, lines denoted interactions, and the size of the nodes represented degrees. The blue genes are reported to be related to neurodevelopmental or psychiatric disorders, the red genes are associated with sleep related traits, the purple genes are associated with ADHD, while gray genes are not reported to be associated with neurodevelopment or sleep. Blue shading: hub genes from the Maximal Clique Centrality (MCC) algorithm.

Following the pleiotropic locus analysis, and through comprehensive literature review and database analysis, we determined that 58 of these genes have established associations with neurodevelopmental disorders, psychiatric conditions, sleep-related phenotypes, or immune system functions (Table S6). Protein-protein interaction network analysis using the STRING database suggested a complex network comprising 97 genes, with several functionally significant modules potentially emerging (Fig. 5D). The primary module (highlighted in blue) comprises the 10 hub genes identified by the Maximal Clique Centrality (MCC) algorithm, which may point to enrichment in growth-related pathways that could imply relevance to neurodevelopmental processes. The gene *NPRL2* encodes a critical component of the GATOR1 complex, which serves as a key negative regulator of the mTOR signaling pathway in the central nervous system. Together with *DEPDC5* and *NPRL3*, *NPRL2* contributes to fundamental neurodevelopmental processes such as neuronal maturation, circuit formation, and synaptic plasticity [55]. Loss-of-function mutations in *NPRL2* are directly linked to Sleep-related Hypermotor Epilepsy, suggesting a potential genetic link between mTOR pathway regulation and the pathophysiology of this sleep-specific neurological disorder [56]. *CACNA2D2*, a candidate gene for neurological disorders including developmental and epileptic encephalopathy, ataxia, and cerebellar atrophy [57], is linked to early infantile epileptic encephalopathy. Given its role in calcium channel function and neuronal excitability, *CACNA2D2* may represent a hypothetical candidate for disrupted synaptic transmission and neural circuit stability that might underlie ADHD and sleep disturbances. *SEMA3F*, a member of the semaphoring III family involved in axon guidance during neuronal development, has been linked to autism [58]. Its role in neural circuit formation raises the hypothesis that it may contribute to ADHD and sleep comorbidity by affecting the development of brain regions involved in attention and circadian regulation. *HYAL1* and *HYAL2* play pivotal roles in nervous system development, neural function maintenance [53, 59], and sleep-related disorders. Of particular relevance to sleep medicine, *HYAL1* is associated with obstructive sleep apnea (OSA) [22, 60, 61]. Together, these genes form an interconnected network that functionally integrates key biological processes—such as neurotransmitter regulation, synaptic function, and neurodevelopment—which are linked to the pathophysiology of ADHD and sleep physiology. Previous research has conceptualized neurodevelopmental disorders (NDDs) as "synaptopathies" or "connectomies" and exploring their relationship with sleep and circadian rhythms [62]. "Synaptopathies" potentially contribute to sleep disturbances [63] and "connectomies" associated with cognitive, behavioral, and sleep-related dysfunctions. These observations generate testable hypotheses for future mechanistic studies.

Our study employed a longitudinal, multi-layered analytical approach. Our phenotypic findings extend previous research [17, 18] by systematically examining this association across multiple time points, subpopulations, and under varying medication statuses. By employing LGCM and incorporating phenotypic factors, we were able to model developmental trajectories while distinguishing between-person differences from within-person changes. This strategy enabled an integrated investigation of ADHD, medication use, and their interaction on sleep traits in the ABCD cohort, providing a potentially useful framework for future longitudinal studies. Identification of within-person lagged effects

highlights potential critical time windows that may inform dynamic intervention, with monitoring of ADHD symptom fluctuations potentially aiding in mitigating subsequent sleep disturbances. Evidence suggests that sleep-targeted interventions improve sleep quality and broader symptom profiles in NDDs [64]. In the genetic component of our study, we identified pleiotropic loci and risk genes associated with both conditions. These findings add to the existing understanding of the genetic architecture link ADHD and sleep disturbances. Collectively, our phenotypic and genetic analyses provide a perspective on the interplay between ADHD and sleep.

Our study has several limitations. First, the predominantly European sample limits generalizability to more diverse populations. Second, while we used the parent-reported SDSC to assess sleep traits as a continuous variable, this approach does not capture clinically defined sleep disorders nor objective sleep parameters, and both ADHD symptoms and sleep problems were derived from parent-reported measures, introducing substantial risk of shared informant bias and shared-method variance that may inflate observed associations. Therefore, future studies should incorporate objective measures such as actigraphy or polysomnography to validate these findings. Third, our post-GWAS analyses combined child sleep phenotypes from the ABCD cohort with adult insomnia GWAS data from the UK Biobank. Although this strategy was employed to enhance power by leveraging large-scale adult sleep GWAS for exploratory pleiotropic mapping, it introduces considerable heterogeneity in age groups, sleep phenotype definitions, and cohort characteristics. Sample size limitations and age differences between datasets may hinder the identification of pleiotropic loci and their biological functions. Accordingly, these cross-cohort genetic results are regarded as strictly exploratory and hypothesis-generating rather than confirmatory evidence. Lastly, the longitudinal analysis is constrained by the short time span and lack of data during adolescence, potentially affecting generalizability. The brief follow-up period fails to capture the full developmental trajectory of ADHD and sleep problems into later adolescence. Missing data across this key maturational stage limits our ability to assess whether observed associations change with age or pubertal development. As a result, the generalizability of our temporal and developmental findings to older adolescent populations remains unclear. These limitations highlight the need for future research with more diverse populations, extended timeframes, and clinical sleep disorder assessments to better understand the ADHD-sleep relationship.

## CONCLUSION

Our findings suggest that ADHD medication mitigates the association between ADHD symptoms and sleep problems. The identified pleiotropic loci offer hypothesis-generating insights into the shared biological underpinnings of this comorbidity, which may provide directions for future research into clinical approaches for comorbid ADHD and sleep disturbances.

## CODE AVAILABILITY

All analyses in this study were performed using publicly available open-source statistical packages in R. No custom in-house code was developed for this research. Version information of all software and packages is provided in the Methods section.

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## ACKNOWLEDGEMENTS

The authors thank the Adolescent Brain Cognitive Development (ABCD) Study (https://abcdstudy.org) and the UK Biobank (https://biobank.ctsu.ox.ac.uk/) consortia for making the data publicly available. We also thank the participants and their families for their contribution to these research projects.

## AUTHOR CONTRIBUTIONS

SC contributed to the research conception, YZ performed the literature search and formal analysis, TP, LL, LC, ZX and LL contributed some data QC and analysis script, SC and WL acquired funding, provided guidance, and reviewed the manuscript. YZ and SC drafted and revised the manuscript. All authors read and approved the final manuscript.

## FUNDING

This work was supported by grants from the National Natural Science Foundation of China (32170613, 82471565 to SC), Clinical Medicine Plus X - Young Scholars Project, Peking University, the Fundamental Research Funds for the Central Universities (to SC), the Innovative Research Team (in Science and Technology) in University of Henan Province (26IRTSTHN017 to WL) and Key Research and Development Projects of Henan Province (241111312800 to WL).

## COMPETING INTERESTS

The authors declare no competing interest.

## ETHICS APPROVAL AND CONSENT TO PARTICIPATE

This study used de-identified data from the Adolescent Brain Cognitive Development (ABCD) Study and the UK Biobank. The ABCD Study was approved by the Institutional Review Board (IRB) at the University of California, San Diego (UCSD). Written informed consent was obtained from a parent or guardian of all participants, and assent was obtained from participants. The UK Biobank study was approved by the UK Biobank Ethics and Governance Committee (EGC). All participants provided written informed consent at enrollment. Owing to the use of previously collected, deidentified, aggregated data, this study did not require institutional review board approval.

## ADDITIONAL INFORMATION

**Supplementary information** The online version contains supplementary material available at https://doi.org/10.1038/s41398-026-04166-4.

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